

MANAGEMENT OF ACUTE CORONARY SYNDROME

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INTRODUCTION

Cardiovascular disease (CVD) is the most common cause of mortality and morbidity worldwide. ACS is often the first clinical manifestation of CVD.

The focus of this guideline is centred on the management of patients who present with chest pain suggestive of ACS in the outpatient setting.

DEFINITION AND CLASSIFICATION

ACS are typically caused by the disruption (rupture or erosion) of an unstable coronary artery atherosclerotic plaque with associated partial or complete coronary artery thrombosis and/or microemboli, resulting in diminished blood flow to the myocardium and subsequent myocardial ischemia.

ACS includes 3 related clinical conditions that exist along a continuum of severity:

1. unstable angina
2. non–ST-segment elevation myocardial infarction (NSTEMI), and
3. STEMI

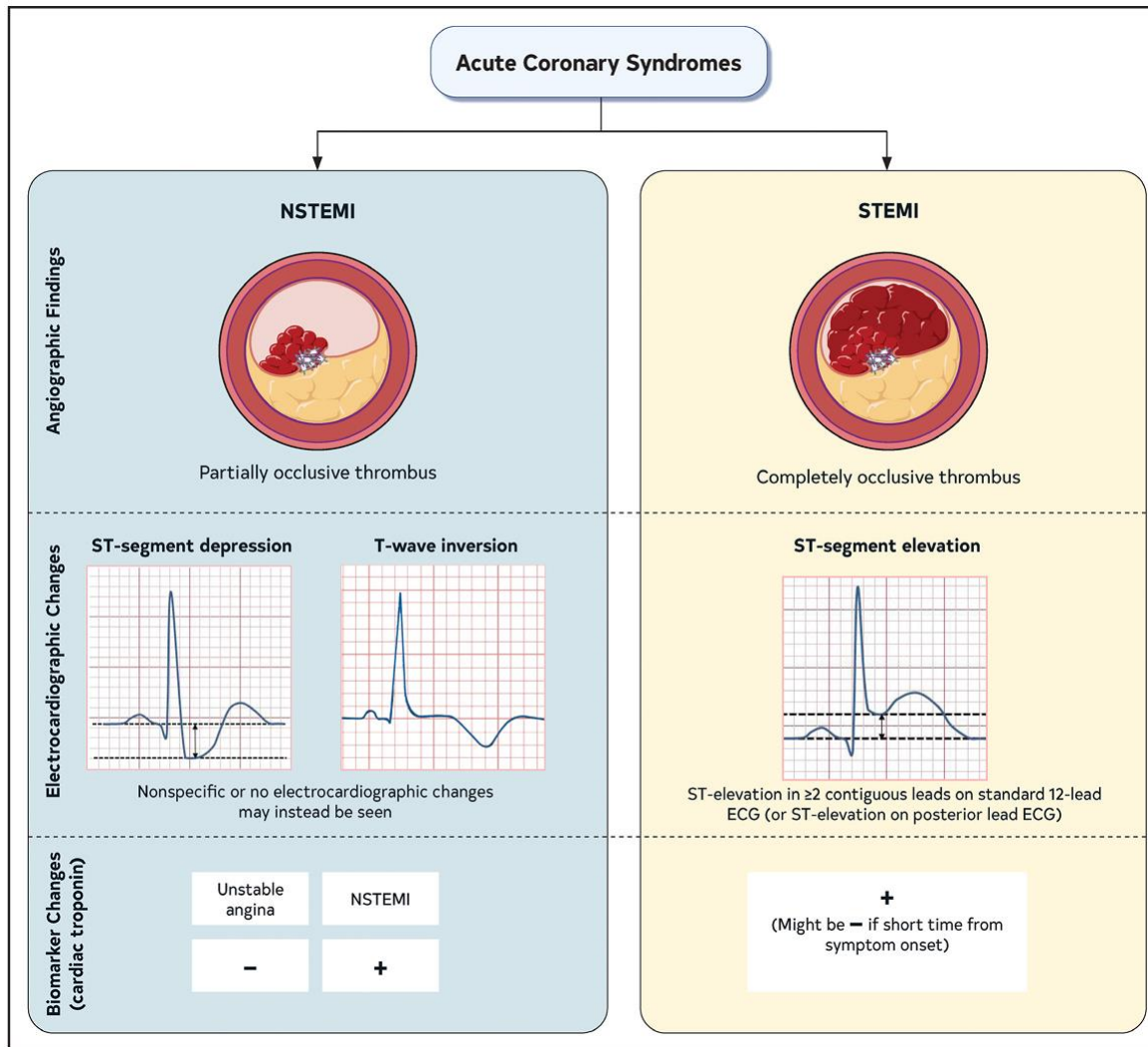
The initial diagnosis and classification of ACS should be based on the:

1. clinical history and symptomatology
2. interpretation of the ECG and
3. assessment of cardiac troponin

Unstable angina is defined by transient myocardial ischemia leading to diminished flow in the absence of significant myonecrosis detected by circulating troponin. It is characterized by specific clinical findings of prolonged (>20 min) angina at rest; new onset of severe angina; angina that is increasing in frequency, longer in duration, or lower in threshold; or angina that occurs after a recent episode of MI.

Patients with NSTEMI may have a partially occluded coronary artery leading to subendocardial ischemia, while those with STEMI typically have a completely occluded vessel leading to transmural myocardial ischemia and infarction. The pathophysiology of ACS can be dynamic and thus patients can rapidly progress from one clinical condition (e.g., unstable angina, NSTEMI, STEMI) to another during the course of their presentation and initial evaluation and treatment.

The figure below shows the types and classification of ACS



(adapted from AHA 2025 guidelines)

DIAGNOSIS

Clinical Presentation

1. Acute chest discomfort—which may be described as pain, pressure, tightness, heaviness, or burning—is the leading presenting symptom prompting consideration of the clinical diagnosis of ACS
2. Chest pain descriptors should be classified as cardiac, possibly cardiac, and likely non-cardiac. The use of the descriptor ‘atypical’ should be avoided, although cardio e-referral still uses atypical.
3. Chest pain-equivalent symptoms include dyspnoea, epigastric pain, and pain in the left or right arm or neck / jaw.

History Taking and Physical Examination

1. Take a focused medical history and accurately characterize the presenting symptoms in order to manage the patient appropriately as soon as possible.
2. Vital signs should be promptly assessed and an initial ECG ordered (these may already have been done by nursing staff in triage room).
3. Physical examination is useful both to eliminate differential diagnoses and to identify very high-risk and high-risk ACS features.
4. Focused physical examination should include checking for the presence of all major pulses, measurement of blood pressure in both arms, auscultation of the heart and lungs, and assessing for signs of HF or circulatory compromise. Pulse and BP discrepancy is a physical sign of aortic dissection.

Differential Diagnoses of Acute Chest Pain

Clinical Syndrome	Findings
Emergency	
Pulmonary Embolism	Tachycardia + dyspnea; pain with inspiration
Aortic dissection	Connective tissue disorders (e.g., Marfan syndrome), extremity pulse differential Severe pain, abrupt onset + pulse differential + widened mediastinum on CXR Syncope
Esophageal rupture	Emesis, subcutaneous emphysema, pneumothorax with unilateral decreased or absent breath sounds

Clinical Syndrome	Findings
Others	
Noncoronary cardiac: Aortic Stenosis, Aortic Regurgitation, Hypertrophic Cardiomyopathy	AS: Characteristic systolic murmur, tardus or parvus carotid pulse AR: Diastolic murmur at right of sternum, rapid carotid upstroke HCM: Increased or displaced left ventricular impulse, prominent <i>a</i> wave in jugular venous pressure, systolic murmur
Pericarditis	Fever, pleuritic chest pain, increased in supine position, friction rub
Myocarditis	Fever, chest pain, heart failure, S3
Esophagitis, peptic ulcer disease, gall bladder disease	Epigastric tenderness Right upper quadrant tenderness, Murphy sign
Pneumonia	Fever, localized chest pain, may be pleuritic, friction rub may be present
Pneumothorax	Dyspnea and pain on inspiration, unilateral absence of breath sounds
Costochondritis, Tietze syndrome	Tenderness of costochondral junctions
Herpes zoster	Pain in dermatomal distribution, triggered by touch; characteristic rash (unilateral and dermatomal distribution)

Distinguishing Cardiac and Non-Cardiac Chest Discomfort and Dyspnoea

		Symptom characteristics	
		↓ Decreasing likelihood of angina/pain from cardiac origin	↑ Increasing likelihood of angina/pain from cardiac origin
Chest discomfort 	Quality	• Burning • Sharp • Tearing – ripping • Pleuritic • Aching	• Strangling • Constricting • Squeezing • Pressure • Heaviness • Associated with autonomic symptoms of heavy perspiration and nausea
	Location and size	• Right • Shifting • Large area or localised to one small spot	• Retrosternal • Extending to left arm, or both arms simultaneously, or to jugular or intrascapular region • 'Fist'-size
	Duration	• Lasting	• Short: up to 5–10 minutes if triggered by physical exertion or emotion
	Trigger	• At rest • On deep inspiration or when coughing • When pressing on ribs or sternum	• On effort • More frequent in cold weather, strong winds or after a heavy meal • Emotional distress (anxiety, anger, excitement or nightmare)
	Relief	• By taking antacid or drinking milk	• Subsiding within 1–5 minutes after effort discontinuation • Relief accelerated by sublingual glyceryl trinitrate
Dyspnoea 	Quality	• Difficulty to exhale • With wheezing	• Difficulty catching breath • Associated with autonomic symptoms of heavy perspiration and nausea
	Trigger	• Both at rest and on effort • While coughing	• On effort
	Relief	• Slowly subsiding at rest or after inhalation of bronchodilators	• Rapidly subsiding after effort discontinuation

(adapted from 2025 ACE Clinical Guidance Management of chronic coronary syndrome)

ECG Interpretation for Patients with Suspected ACS

	Non-ST Elevation Acute Coronary Syndrome (NSTEMI-ACS)	ST Elevation Myocardial Infarction (STEMI)
Electrocardiographic evidence of ischemia	New or presumed new and usually dynamic horizontal or down-sloping ST-segment depression ≥ 0.5 mm in ≥ 2 contiguous leads and/or T-wave inversion >1 mm in ≥ 2 contiguous leads with prominent R wave or R/S ratio >1 or transient ST-segment elevation.	New or presumed new ST-elevation of ≥ 1 mm in ≥ 2 anatomically contiguous leads (measured at the J-point) in all leads other than V2-V3 and ≥ 2 mm in men ≥ 40 y, ≥ 2.5 mm in men <40 y, and ≥ 1.5 mm in women regardless of age in leads V2-V3.*
Other observed electrocardiographic changes	Many patients with NSTEMI-ACS have either nonspecific ST-segment or T-wave changes or a normal ECG. The absence of electrocardiographic evidence of ischemia does not exclude ACS.	Posterior leads (V7-V9) should be obtained in patients with suspected left circumflex occlusion particularly in the setting of isolated ST-segment depression ≥ 0.5 mm in leads V1-V3.

* ST-segment changes may be observed in other conditions including acute pericarditis, LVH, LBBB, Brugada syndrome, right ventricular pacing, Takotsubo syndrome, and early repolarization that may obscure the diagnosis of STEMI. New or presumably new LBBB at presentation occurs infrequently and should not be considered diagnostic of AMI in isolation; clinical correlation is required. A new LBBB in an asymptomatic patient does not constitute a STEMI equivalent.

MANAGEMENT AND FOLLOW-UP

Antiplatelet Therapy

- Give aspirin loading dose 300mg orally if not contraindicated
- Aspirin (nonenteric coated) should be chewed, when possible, to achieve faster onset of antiplatelet action
- Loading dose should be administered for patients who are already on aspirin therapy

Acute Management in The Polyclinic

- Transfer patient to the triage/treatment room
- Inform nursing staff to call for the ambulance

Oxygen Therapy

- Oxygen supplementation is recommended in ACS patients with hypoxaemia (oxygen saturations < 90%)
- Oxygen supplementation in patients who are not hypoxic (oxygen saturations > 90%) is not associated with clinical benefits and is therefore not recommended

Analgesia

- Patients presenting with known or suspected ACS often experience chest pain or other uncomfortable symptoms. Rapid and effective pain relief remains an important treatment goal to prevent sympathetic activation and adverse clinical sequelae
- Sublingual GTN may be given but only in hemodynamically stable patients with SBP $\geq$ 90mmHg
- Nitrates should not be administered after recent PDE5 inhibitor use. Avoid use of nitrates / nitroglycerin within 12 hours of avanafil, 24 hours of sildenafil / vardenafil, or 48 hours of tadalafil.

REFERRALS TO RAPID ACCESS CHEST PAIN CLINIC

- Patients with chest pain that is suggestive of ischaemia but are assessed to be stable and do not have ACS can be referred to the Rapid Access Chest Pain Clinic (RACPC)
- Open to all NUP Clinics to refer
- Clinic runs on Mon- Fri only at NTFGH
- Appointments will generally be given for patient to be seen within 2-3 working days*
- Appointment will be paired** with a Treadmill Exercise ECG test hence patient must be able to a TMX test

* Patients who are referred on a Thursday / Friday may only be seen early in the following week

** A blood test appointment may also be made for the visit date

RACPC Inclusion and Exclusion Criteria

Inclusion Criteria	Exclusion Criteria
• Age $\geq$ 18 years old • Episode (s) of Chest Pain (assessed <u>unlikely</u> to be musculoskeletal, pleuritic or gastric in origin) • Baseline ECG done on referral day • Has CV risk factors based on Age / Gender / DM / Hypertension / Hyperlipidaemia / Smoking / Ethnicity • CAD history: ○ No known history of CAD <u>or</u> ○ History of CAD more than 1 year prior to referral date with no existing Cardiology SOC following up	• Suspected Cardiac emergencies e.g. suspected acute MI, severe / acute heart failure, unstable angina (<i>refer ED</i>) • ECG abnormality as below (<i>refer ED or Gen Cardio SOC</i>) e.g.: ○ Complete LBBB ○ $> 1\text{mm}$ resting ST depression ○ Tachy/Bradyarrhythmias • Severe arterial hypertension (SBP $> 200\text{mmHg}$ and/or DBP $> 110\text{mmHg}$) • On digoxin • Has existing / upcoming appointment with Cardiology SOC in any PHI (<i>refer back to relevant Cardio SOC</i>) • Known to have normal CTCA / invasive coronary angiogram in the past 1 year (<i>refer Cardio SOC if necessary</i>)

Workflow for doctors when referring to RACPC

- To order ECG for every patient being referred to RACPC
- If ECG was done in triage room to ensure the ECG is uploaded in Epic
- Place the correct referral order – Referral to Cardiology – Rapid Access Chest Pain Clinic – NTFGH
- Order TCU in 2 weeks – communicate to patient that the purpose is to review results of RACPC and to review chest pain symptom
- Send patient to NUP referral counter for the RACPC appointment to be made

REFERENCES

1. 2025 ACC/AHA/ACEP/NAEMSP/SCAI Guideline for the Management of Patients with Acute Coronary Syndromes
2. 2023 ESC Guidelines for the management of acute coronary syndromes
3. SMA Medical Guidelines on Fitness to Drive, 2nd Edition, 2011